

Nathaniel Fransiscus Mapaye

nmapaye@ucsc.edu | (831) 227-4349 | linkedin.com/in/nmapaye | github.com/nmapaye | nmapaye.com

EDUCATION

University of California, Santa Cruz

B.S., Technology & Information Management

Undergraduate Dean's Scholarship Award | Bilingual International Baccalaureate Diploma

Expected Jun 2027

Santa Cruz, CA

EXPERIENCE

Frontier AI Lab | Large Language Model (LLM) Specialist

Jun 2026 - Aug 2026

San Francisco, CA | Contract

- Evaluated 520+ frontier AI outputs across software engineering and technical reasoning against 3 criteria (correctness, instruction following, and edge cases) to produce reproducible evidence for model-improvement workflows.
- Diagnosed recurring 70+ failure patterns and generated evidence-backed annotations, giving reviewers a clear, reproducible basis for model-improvement reviews, 89 percent of them could be re-run with the same behavior.

CFX Indonesia | Cyber Security AI Red-Teamer Intern

Jun 2026 - Present

Jakarta, Indonesia

- Assessed 30+ assets across web, network, and control workflows and security risks across 3 workstreams (risk analysis, control documentation, and vulnerability remediation), producing traceable security evidence for an OJK-regulated digital-asset exchange.
- Analyzed and triaged 27+ vulnerabilities in regulated market infrastructure and tracked findings through remediation, maintaining a clear evidence trail under program constraints.

Handshake | Subject Matter Expert (SME) - AI Alignment

Apr 2026 - Present

Remote | Contract

- Audited 8+ repository-level software tasks across 4 evidence sources (code, diffs, tests, and requirements), documenting reproducible failure modes and implementation gaps under structured rubrics.

Bitera DC | Data Center Engineering Intern

Aug 2025 - Sep 2025

Jakarta, Indonesia

- Verified 16+ high-severity web findings through authenticated testing and retesting, producing actionable remediation evidence while supporting 5 live data-center domains (Linux, networks, power, cooling, and alarms).

Xapiens Teknologi Indonesia | Cyber Security Analyst Intern

Jul 2024 - Sep 2024

Tangerang, Indonesia

- Executed authenticated scans and manual web tests on 50+ endpoints, validated exploitability, and produced proof-of-concept reports that prioritized remediation for several CVSS 8.0+ findings.

PROJECTS

AURORA | React Native Caffeine and Alertness Tracker

Aug 2025 - Present

iOS, TypeScript/JavaScript, Swift, Apple HealthKit, SQLite

- Built a React Native iOS app with Swift and Apple HealthKit that unified 3 workflows (caffeine logging, intake dashboards, and psychomotor-vigilance testing).
- Engineered 2-layer local-first storage with AsyncStorage and SQLite, normalized caffeine inputs to milligrams, and unified data across 2 logging paths (automatic and manual).

EmbNode | FreeRTOS Embedded Telemetry Node

Aug 2025 - Sep 2025

C++23, FreeRTOS, ESP32/STM32, CMake/CTest

- Designed a C++23/FreeRTOS DMA pipeline across 3 telemetry stages (high-priority sampling, aggregation, and MQTT/HTTP delivery), with OTA and watchdog recovery for resilient low-power operation.
- Validated CRC16-CCITT and telemetry decoding in a 0.47-second CMake/CTest host simulation, verifying data integrity while modeling 0.15 mA average current on a 15-second sleep schedule.

TECHNICAL SKILLS

Languages: C/C++23, Python, Swift, Go, TypeScript/JavaScript, Java, Ruby, CSS, HTML, Astro, HCL, Lua

Mobile and Systems: iOS, React Native, Apple HealthKit, FreeRTOS, ESP32/STM32, Linux/Unix, SQLite, concurrency, data structures and algorithms, object-oriented design, debugging

APIs and Infrastructure: REST, MQTT, client-server design, Git/GitHub, Docker, Kubernetes, GitHub Actions

Information Security and Testing: Burp Suite, Nessus, OWASP Testing Guide, OWASP ASVS, CVSS, Ghidra, CMake/CTest